The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 November 1999.

INCH-POUND

MIL-PRF-19500/485F
30 August 1999
SUPERSEDING
MIL-S-19500/485E
15 November 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPES: 2N5415, 2N5415S, 2N5416, AND 2N5416S
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See figure 1 (similar to T0- 5), and 2 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_{OP}	$R_{\theta JC}$
	W	W	V dc	V dc min	V dc	A dc	°C	°C/W
2N5415, S	0.75	10	200	200	6.0	1.0	-65 to +200	17.5
2N5416, S	0.75	10	350	300	6.0	1.0	-65 to +200	17.5

1/ Derate linearly 4.28 mW/°C for T_A > +25°C.

2/ Derate linearly 57.1 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics.

	h_{FE1} I_C = 50 mA dc, 1/ V_{CE} = 10 V dc	C_{obo} I_E = 0, V_{CB} = 10 V dc, 100 kHz ≤ f ≤ 1 MHz	hfe I_C = 10 mA dc V_{CE} = 10 V dc, f = 5 MHz	V_{BE} I_C = 50 mA dc, 1/ V_{CE} = 10 V dc	$V_{CE(sat)}$ I_C = 50 mA dc, 1/ I_B = 5 mA dc
		pF		V dc	V dc
Min	30		3		
Max	120	15	15	1.5	2.0

1/ Pulsed (See 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

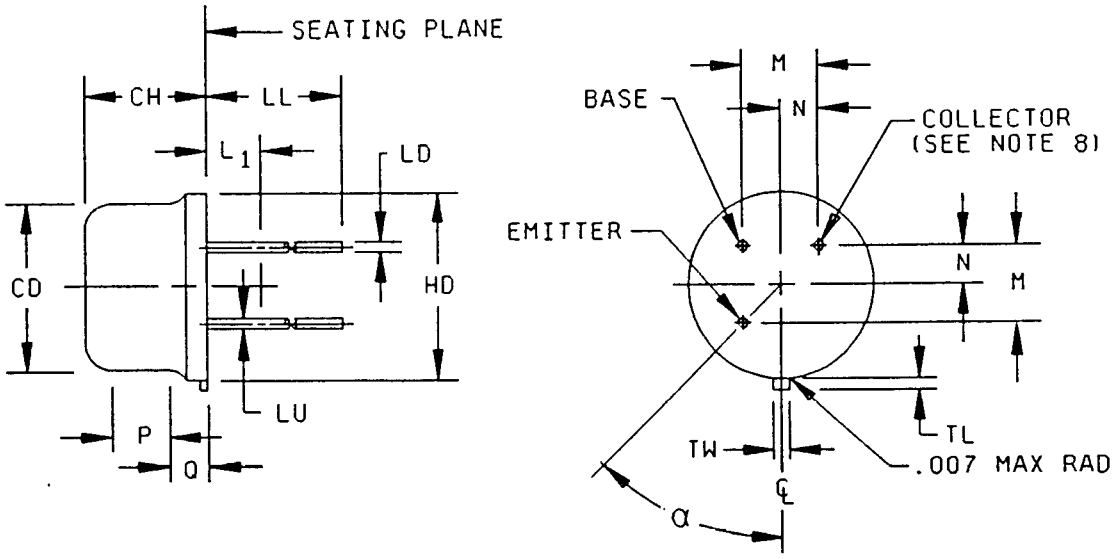
MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				Notes		Ltr	Dimensions				Notes
	Inches		Millimeters					Inches		Millimeters		
	Min	Max	Min	Max				Min	Max	Min	Max	
HD	.335	.370	8.51	9.40			P	.100		2.54		5
CD	.305	.335	7.75	8.51			Q					6
CH	.240	.260	6.10	6.60			TL	.029	.045	0.74	1.14	9
LL	See notes				10,12,13		TW	.028	.034	0.71	0.86	
L1		.050		1.27	11		a	45° TP				7
LD	.016	.021	0.41	0.53	3,10		M	.1414 Nom		3.59 Nom		7
LU	.016	.019	0.41	0.48	4,10		N	.0707 Nom		1.80 Nom		7

FIGURE 1. Physical dimensions (similar to T0- 5).

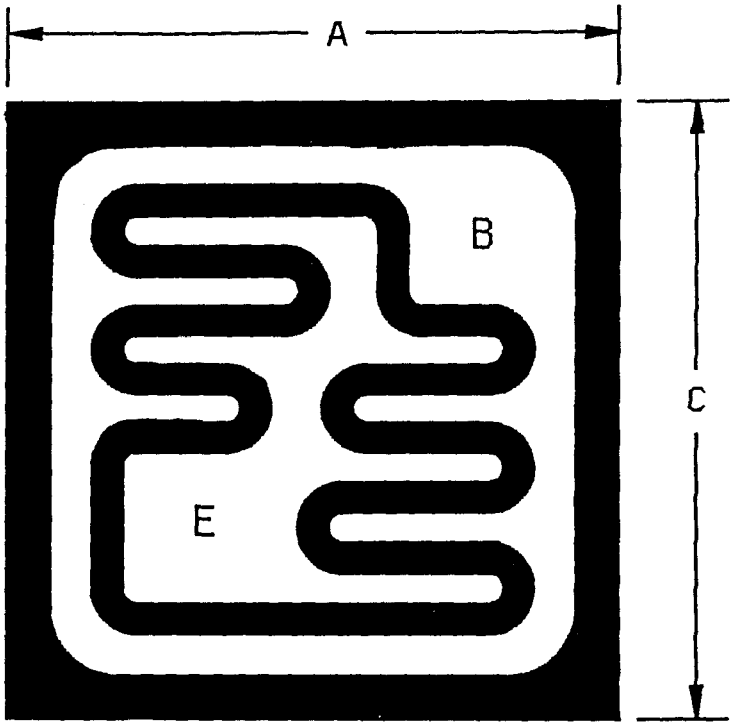
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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .050 inch (1.27 mm) and .250 inch (6.35 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed .010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in gauging plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred method of measurement.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All 3 leads (see 3.3.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N5415 and 2N5416, dimension LL shall be 1.500 inches (38.10 mm) minimum and 1.75 inches (44.5 mm) maximum.
13. For transistor types 2N5415S and 2N5416S, dimension LL shall be 0.5 inch (12.70 mm) minimum and 0.75 inch (19.05 mm) maximum.
14. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions - Continued.

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A version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	0.99	1.09
C	.039	.043	0.99	1.09

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
3. The physical characteristics of the die are:
   Thickness is .006 inch (0.15 mm) minimum, .012 inch (0.30 mm) maximum.
   Top metal: Aluminum 17,500 Å nominal, 20,000 Å nominal
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector; Bonding pad: B = .004 inch (0.10 mm) x .005 inch (0.13 mm).
   E = .004 inch (0.10 mm) x .0055 inch (0.14 mm).

FIGURE 2. JANHC and JANKC (A-version) die dimensions.

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3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2 herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC die. Qualification for die shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{CEX}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CEX} , h_{FE1} ΔI_{CEX} = 100 percent of initial value or 500 nA dc, whichever is greater Δh_{FE1} = ± 15 percent of initial value	I_{CEO1} and h_{FE1}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX} = 100 percent of initial value or 500 nA dc, whichever is greater; Δh_{FE1} = ±15 percent of initial value	Subgroup 2 of table I herein; ΔI_{CE01} = 100 percent of initial value or 500 nA dc, whichever is greater; Δh_{FE1} = ± 20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5:

$V_{CB} \geq 35$ V dc, $P_T = 700$ mW .

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ---------- 10 mA min.
b. I_H forward heating current ---------- 200 mA to 330 mA.
c. t_H heating time ---------- 50 - 100 ms.
d. t_{md} measurement delay time ---------- 60 µs max.
e. V_{CE} collector-emitter voltage ---------- 10 V dc minimum
f. V_{CB} collector-base voltage ---------- 10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 75 °C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 35 V dc; 2,000 cycles.
B5	1027	V_{CB} = 35 V dc; P_T adjusted as required to achieve T_J and time of MIL-PRF-19500.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 - 30 V dc, adjust P_T to achieve T_J = 150°C min. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, c = 0
2	1039	The steady state life test of step 1 shall be extended to 1,000 hrs for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-Temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	T_A = +30°C ±5°C; $V_{CB} \geq$ 100 V dc; adjust P_T to achieve TJ = 150°C min.

4.4.3.1 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance for JAN, JANTX, and JANTXV. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following conditions shall apply: Max $R_{\theta JC}$ shall be 17.5°C/W.

a. I_M: Measurement current ..... 10 mA dc.
b. V_{CE}: Measurement current (same as V_H) ..... 30 V dc.
c. I_H: Collector heating current ..... 0.33 A.
d. V_H: Collector-emitter heating voltage ..... 30 V dc
e. t_H: Heating time ..... 1 second minimum
f. t_{MD}: Measurement delay time ..... 50 µs
g. t_{SW}: Sampling window time ..... 10 µs maximum

4.5.3 Thermal resistance for JANS. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following conditions shall apply: Max $R_{\theta JC}$ shall be 17.5°C/W.

a. I_M: Measurement current ..... 10 mA dc.

b. V_{CE}: Measurement current (same as V_H) ..... 25 V dc.

c. I_H: Collector heating current ..... 0.2 A.

d. V_H: Collector-emitter heating voltage ..... 25 V dc

e. t_H: Heating time ..... 1 second minimum

f. t_{MD}: Measurement delay time ..... 50 µs

g. t_{SW}: Sampling window time ..... 10 µs maximum

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/, 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/, 4/, 5/	1022	n = 15 devices, c = 0				
Temp Cycling 3/, 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic Seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Group A, subgroup 2				
Bond strength 3/, 5/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Collector to emitter cutoff current	3041	Bias condition D	I_{CE01}		50	µA dc
2N5415,S 2N5416,S		V_{CE} = 150 V dc V_{CE} = 250 V dc				
Collector to emitter cutoff current	3041	Bias condition D	I_{CE02}		1	mA dc
2N5415,S 2N5416,S		V_{CE} = 200 V dc V_{CE} = 300 V dc				
Collector to emitter cutoff current	3041	Bias condition D, V_{BE} = 1.5 V dc	I_{CEX}		50	µA dc
2N5415,S 2N5416,S		V_{CE} = 200 V dc V_{CE} = 300 V dc				
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		20	µA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		50	µA dc
2N5415,S 2N5416,S		V_{CB} = 175 V dc V_{CB} = 280 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}		500	µA dc
2N5415,S 2N5416,S		V_{CB} = 200 V dc V_{CB} = 350 V dc				

See footnote at end of table

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup2 - continued						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 50 mA dc, Pulsed (see 4.5.1)	h_{FE1}	30	120	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1 mA dc, Pulsed (see 4.5.1)	h_{FE2}	15		
Collector to emitter voltage (saturated)	3071	I_C = 50 mA dc, I_B = 5 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)}$		2.0	V dc
Base emitter voltage saturation	3066	Test condition B, V_{CE} = 10 V dc, I_C = 50 mA dc, Pulsed (see 4.5.1)	V_{BE}		1.5	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D	I_{CB03}		1	mA dc
2N5415,S 2N5416,S		V_{CB} = 175 V dc V_{CE} = 280 V dc				
Low temperature operation:		T_A = -65°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 50 mA dc, Pulsed (see 4.5.1)	h_{FE3}	15		
Subgroup 4						
Small-signal short- circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 10 mA dc, f = 5 MHz	$\lvert h_{fe} \rvert$	3	15	
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 5 mA dc, f ≤ 1 kHz	h_{fe}	25		
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		15	pF
Input capacitance(output open-circuited)	3240	V_{EB} = 5 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		75	pF
Subgroup 5						
Pulse response	3251	Test condition A				
Turn-on time		V_{CC} = 200 V dc, I_C = 50 mA dc, I_{B1} = 5 mA dc, see figure 3	t_{on}		1	µs
Turn-off time		V_{CC} = 200 V dc, I_C = 50 mA dc, I_{B1} = I_{B2} = 5 mA dc, see figure 3	t_{off}		10	µs

See footnote at end of table.

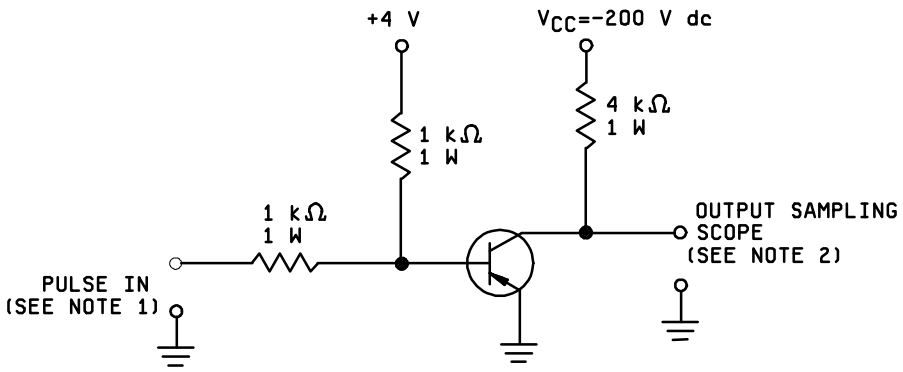
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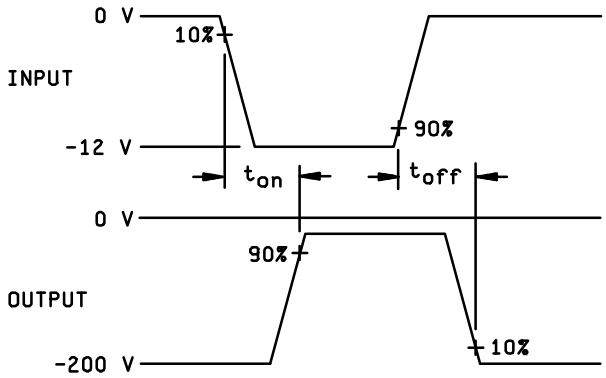
TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Not applicable						
Subgroup 7 4/						
Breakdown voltage, collector to emitter	3011	I_C = 50 mA dc, I_B = 5 mA dc, L = 25 mH; f = 30 - 60 Hz (see figure 4)	$V_{(BR)CEO}$			
2N5415, S				200		V dc
2N5416, S				300		V dc
Safe operating area (continuous dc)	3051	T_C = +25°C, 1 cycle, t = 0.4 s, (see figure 5). n = 22, c = 0				
Test 1		V_{CE} = 10 V dc, I_C = 1 A dc				
Test 2		V_{CE} = 100 V dc, I_C = 100 mA dc				
Test 3 (2N5415, S only)		V_{CE} = 200 V dc, I_C = 24 mA dc				
Test 4 (2N5416, S only)		V_{CE} = 300 V dc, I_C = 10 mA dc				
Electrical measurements		See table I, group A, subgroup 2 herein.				

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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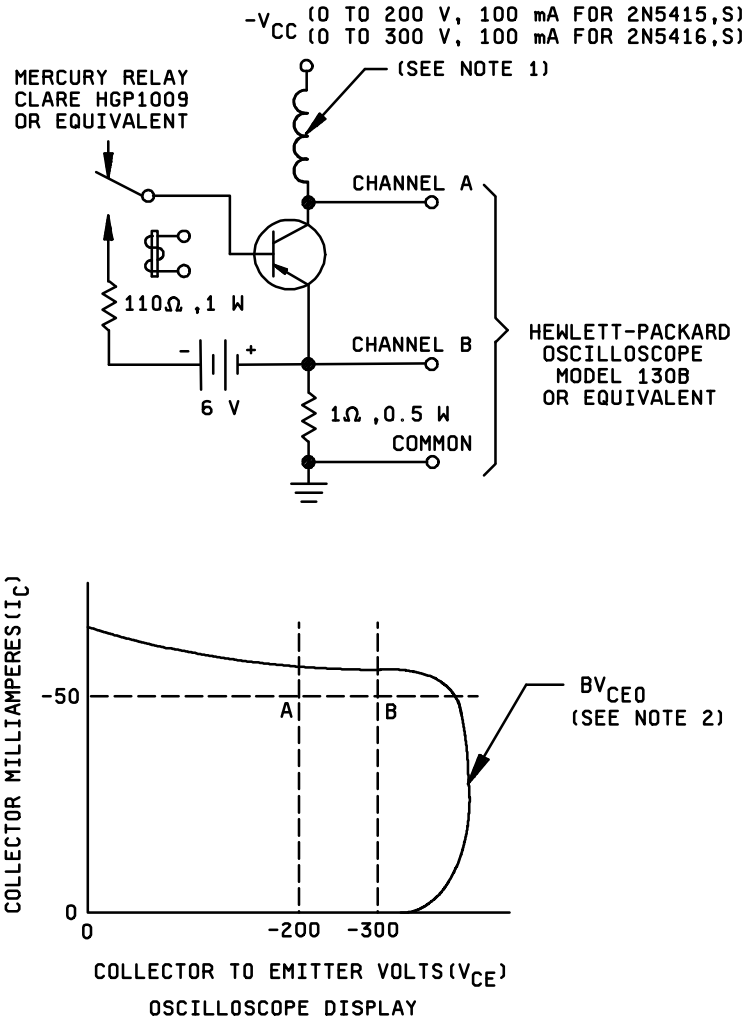


NOTES:

1. The rise time (t_r) and fall time (t_f) of applied pulse shall be each < 20 ns; duty cycle < 2 percent; generator source impedance shall be 50 Ω; pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{in} > 100 kΩ; C_{in} < 50 pF; rise time < 2 ns.

FIGURE 3. Pulse response test circuit.

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NOTES:

1. L = 25 mH, Q = 76, R = 83.4 Ω, I = 75 mA (J.W. Miller number 6308 or equivalent).
2. BV_{CEO} is acceptable when the trace falls to the right and above point “A” for type 2N5415. The trace shall fall to the right and

FIGURE 4. BVCEO measurement test circuit.

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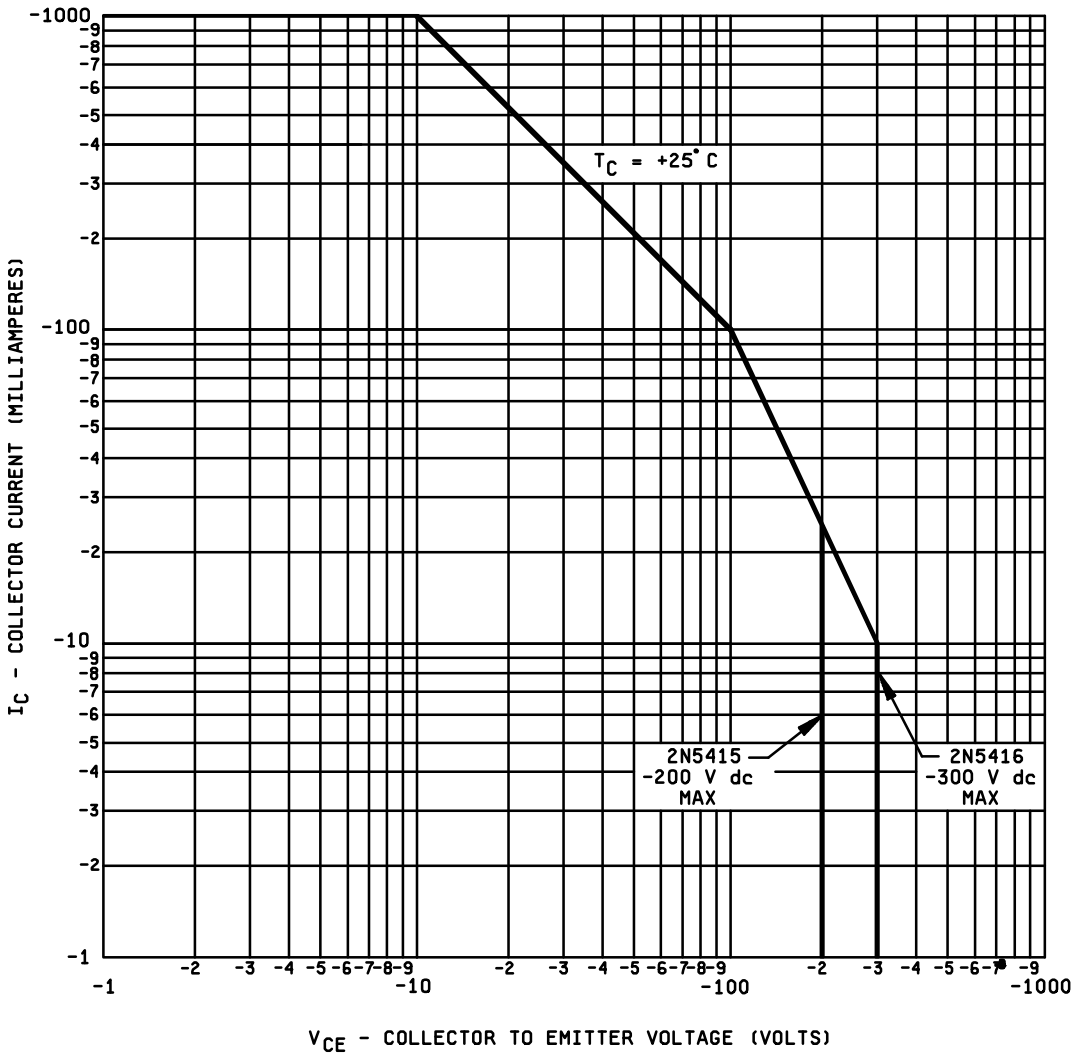


FIGURE 5. Maximum safe operating graph (continuous dc).

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.4.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML 19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA2N5415) will be identified on the QPL.

JANC ordering information	
PIN	Manufacturers CAGE 33178
2N5415	JANHCA2N5415 JANKCA2N5415
2N5416	JANHCA2N5416 JANKCA2N5416

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

(Project 5961-2038)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/485F	**2. DOCUMENT DATE (YYMMDD)** 99/08/30

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON LOW-POWER, TYPES 2N5415, 2N5415S, 2N5416, AND 2N5416S, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD FORM 1426, FEB 99 (EG) PREVIOUS EDITIONS ARE OBSOLETE WHS/DIOR, Feb 99

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